

Pokhara University
Faculty of Science and Technology

Course Code.: CMP XXX (3 Credits)
Course Title: Computer Vision (3-0-0)
Nature of the course: Theory + Practical
Level: Master

Full marks: 100
Pass marks: 60
Total Lectures: 45 hrs
Program: MCE/MSc CS

1. Course Description

This course provides an in-depth study of computer vision, emphasizing modern machine learning and deep learning techniques for visual data analysis. Students will explore algorithms for image recognition, object detection, image segmentation, and image generation, along with their real-world applications. The course also introduces recent research trends and practical implementations of computer vision systems.

2. General Objectives

The course is designed with the following general objectives:

- Understand advanced concepts and techniques in computer vision.
- Apply machine learning and deep learning methods to visual data analysis.
- Analyze real-world visual datasets and develop intelligent vision-based systems.
- Explore current research trends and applications in computer vision.
- Develop practical applications using modern computer vision techniques.

3. Methods of Instruction

Lecture, Discussion, Blended, Case Studies, Project and Research Works.

4. Contents in Detail

Specific Objectives	Contents
• Understand the fundamental concepts and the image processing pipeline used in computer vision systems.	1. Foundations of Computer Vision (4 hrs) 1.1. Overview of computer vision and its applications 1.2. Image acquisition and preprocessing techniques 1.3. Image augmentation and feature extraction 1.4. Spatial filtering and edge detection methods Case Study: Image acquisition, augmentation, normalization, and feature representation for human face recognition systems.
• Understand and apply deep neural network architectures for visual recognition and analysis tasks.	2. Deep Learning for Computer Vision (6 hrs) 2.1. Convolutional Neural Networks (CNN) 2.2. Deep neural network training and optimization 2.3. Regularization techniques and hyperparameter tuning 2.4. Domain adaptation and transfer learning Case Study: Visual feature representation and automation systems for animal classification.
• Learn advanced techniques for detecting and recognizing multiple objects in images.	3. Object Detection and Recognition (8 hrs) 3.1. Concepts of object detection 3.2. Classical object detection methods 3.3. Two-stage detectors: R-CNN, Fast R-CNN, Faster R-CNN, RPN, Mask R-CNN 3.4. One-stage detectors: YOLO family, SSD

	3.5. Evaluation metrics: Precision, Recall, Mean Average Precision (mAP), Intersection over Union (IoU) Case Study: Detecting objects in self-driving car systems.
Understand segmentation techniques for detailed image and scene understanding.	4. Image Segmentation (9 hrs) 4.1. Concept and importance of image segmentation 4.2. Types of segmentation: Semantic, Instance, and Panoptic segmentation 4.3. Classical segmentation methods: Thresholding, Region Growing, Watershed algorithm 4.4. Evaluation metrics: IoU, Dice coefficient, Pixel accuracy 4.5. Deep learning models: U-Net architecture and DeepLab networks Case Study: Medical image segmentation for detecting tumors or organs using deep neural networks.
Learn techniques for estimating motion and tracking objects across video sequences for dynamic scene analysis.	5. Object Motion and Tracking (10 hrs) 5.1. Motion estimation and optical flow (Lucas-Kanade and Horn-Schunck methods) 5.2. Background subtraction and foreground/moving object detection 5.3. Classical tracking methods: Kalman Filter, Particle Filter, Mean-Shift/CAMShift 5.4. Deep learning-based tracking: Siamese networks, correlation filters, SORT and DeepSORT 5.5. Multi-object tracking and evaluation metrics (MOTA, MOTP, IDF1) Case Study: Real-time vehicle and pedestrian tracking for traffic monitoring and surveillance systems.
Explore real-world applications and emerging areas of computer vision technologies.	6. Applications of Computer Vision (8 hrs) 6.1. Computer vision in public service delivery 6.2. Intelligent transportation systems 6.3. Medical image analysis 6.4. Climate change monitoring and disaster management 6.5. Applications in agriculture and tourism support systems Case Study: Precision agriculture using computer vision to monitor crop health and detect plant diseases.

5. Evaluation System and Students' Responsibilities

Evaluation System

The internal evaluation of a student may consist of assignments, attendance, internal assessment, project and research works etc. The internal evaluation scheme for this course is as follows (but not limited to):

Internal Evaluation	Weight	Marks	External Evaluation	Marks
Attendance & Class Participation	10%	60	Semester-End examination	40
Assignments	20%			
Internal Assessment	40%			
Research/Project/Case Study Work	30%			

Total Internal		60	Total External	40
Full Marks: 60 + 40 = 100				

Students' Responsibilities

Each student must secure at least 60% marks separately in internal evaluation with 80% attendance in the class in order to appear in the Semester End Examination. Failing to get such a score will be given NOT QUALIFIED (NQ) to appear for the Semester-End Examinations. Students are advised to attend all the classes, formal exam, test, etc. and complete all the assignments within the specified time period. Students are required to complete all the requirements defined for the completion of the course.

6. Reference Books

1. Szeliski, R. (2022). Computer vision: algorithms and applications. Springer Nature. online at: <http://szeliski.org/Book/>
2. Brownlee J. (2019). *Deep Learning for Computer Vision: Image Classification, Object Detection, and Face Recognition in Python*, Machine Learning Mastery.
3. Siddiqui M.A. (2025), Mastering Computer Vision with PyTorch 2.0, Orange Education Pvt Ltd, AVA
4. Buduma N. et al. (2022). *Fundamentals of Deep Learning Designing Next-Generation Machine Intelligence Algorithms*, O'Reilly Media, Inc.
5. Atienza R. (2020). *Advanced Deep Learning with TensorFlow 2 and Keras*, Packt Publishing Ltd.
6. Michelucci U. (2022). *Applied Deep Learning with TensorFlow 2: Learn to Implement Advanced Deep Learning Techniques with Python*, Apress.
7. Long L., and Zeng X.(2022). *Beginning Deep Learning with TensorFlow: Work with Keras, MNIST Data Sets, and Advanced Neural Networks*, Apress.
8. Bhowmik M.K. (2024). *Computer Vision: Object Detection in Adversarial Vision*, CRC Press.
9. Gallatin K. and Albon C. (2023) Machine Learning with Python Cookbook: Practical Solutions from Preprocessing to Deep Learning, O'Reilly Media, Inc.